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Research paper

Towards resilient energy decisions: Integrating scenario-based weighting and Monte Carlo sensitivity in hybrid renewable system selection in Algeria

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ABSTRACT

Given the challenges of ensuring sustainable energy supply across different regions of Algeria and the need to reduce reliance on fossil fuels, this study presents a comprehensive framework for evaluating and ranking hybrid renewable energy systems. In two climatically and resource-wise distinct regions (Tamanrasset and Adrar), techno-economic simulations were conducted using HOMER software, and multi-criteria decision-making (MCDM) analysis was performed based on the SAW and TOPSIS methods. Seven key criteria—including initial cost, levelized cost of electricity (LCOE), payback period, renewable energy production, energy losses, CO_2 emissions, and population coverage—were evaluated. The results showed that the Tamanrasset option, with a fully solar-based structure, zero CO_2 emissions, an LCOE of \$0.225 per kilowatt-hour, and a payback period of 0.29 years, ranked first in five out of six defined scenarios. In contrast, the Adrar option, based on a wind/diesel structure with annual production of 7425 kWh, ranked first only in the performance-focused scenario. Monte Carlo sensitivity analysis with 5000 iterations revealed that Tamanrasset remained the top choice in over 99 % of cases. The proposed framework, by integrating scenario analysis, uncertainty assessment, and multi-criteria modeling, serves as an effective tool for decision-making in clean energy transition projects.

1. Introduction

In Algeria, with its diverse climates, abundant resources, and favorable geography, developing solar and wind energy is not a choice but a necessity [1]. The historical dependency on fossil fuels, fluctuations in global energy prices, and international pressures to reduce greenhouse gas emissions compel countries such as Algeria to move towards more sustainable energy models [2,3]. On the other hand, population growth, the increasing demand for electricity in remote areas, and the need to reduce reliance on fuel imports further emphasize the importance of utilizing renewable resources at the national level. Accurately identifying the technical potential of the best solar and wind stations is crucial for national planning. Knowing the real production capacity at these sites supports accurate investment estimation, infrastructure design, and regional policy development [4,5]. In other words, accurately estimating the production capacity of renewable resources at key points offers a scientific and practical basis for transforming the idea of clean energy into a practical reality.

Currently, only about 3 % of Algeria's electricity consumption (approximately 686 MW) is supplied from renewable sources such as solar (448 MW), hydropower (228 MW), and wind (10 MW) [6]. Given that Algeria's economy heavily depends on fossil fuel exports, domestic use of these resources could reduce the country's foreign exchange earnings. Therefore, experts warn that if Algeria does not significantly develop its renewable resources by 2030, it will be forced to halt part of its oil and gas exports to meet domestic electricity needs. Fig. 1 shows Algeria's targeted renewable energy mix by 2030 [7]. The largest share is allocated to photovoltaic solar energy with a capacity of 13,575 MW, highlighting the crucial role of this source in the country's energy future. After that, wind energy with 5010 MW plays a significant complementary role. Solar thermal energy occupies the third place with 2000 MW. Other sources include biomass with 1000 MW, combined heat and power with 400 MW, and finally geothermal energy with 15 MW. This figure shows the focus of Algeria's renewable energy policies on solar and wind resources and also highlights the relative importance of other technologies. Information, including the type of energy and the associated capacity, is very useful for energy sector decision-makers and

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Nomenclature

i	Annual interest rate (%)
A	Rotor swept area (m^2)
v	Wind speed (m/s)
C_p	Turbine power coefficient (-)
$E_{load\ served}$	Real electrical load by system (kWh/yr)
$P_{Batt,\ max}$	Maximum battery charge power (kWh)
$P_{Batt,\ kbm}$	Maximum battery charge power base on kinetic battery model (kWh)
$P_{Batt,\ mcr}$	Maximum battery charge power base on maximum charge rate (kWh)
$P_{Batt,\ mcc}$	Maximum battery charge power base on maximum charge current (kWh)
Y_{PV}	Output power of solar cell under standard conditions (kW)
f_{PV}	Derating factor (%)
$\overline{H_T}$	Incident radiation on the cell's surface on a monthly basis (kW/m^2)
$\overline{H_{T,STC}}$	Incident radiation on the cell's surface under standard conditions (1 kW/m^2)
P_{PV}	Output power of PV cells (kW)
ρ_0	Air density at standard pressure and temperature equal to 1.225 kg/m^3
$C_{ann,total}$	Total annual cost ($)
μ	Mean value of outputs (-)
S_j	Sum of the weighted normalized (-)
v_i^-	Maximum of v_{ij} for non-beneficial criterions; minimum of v_{ij} for beneficial criterions
$\widetilde{r}_{ij}$	Normalized i th criterion value for j-th object (-)
C_j^*	Performance score (-)
x_{ij}	The value of criterion *j* for alternative *i* (-)
ω_i	Weighted of i th criterion (-)
ρ	Actual air density (kg/m^3)
η_{gen}	Electrical efficiency of generator (%)
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution (-)
SAW	Simple Additive Weighting (-)
P_{WTG}	Power output of wind turbine (kW)
$P_{WTG,\ STP}$	Power output of wind turbine at standard pressure and temperature (kW)
PV	Photovoltaic (-)
WT	Wind turbine (-)
LCOE	Levelized cost of electricity ($/kWh)
$\eta_{batt,\ c}$	Batteries charge efficiency (%)
N	Useful life-time (year)
η_{Conv}	Converter charge efficiency (%)
NPC	Net present cost ($)
MCDM	Multi-criteria decision-making (-)
n	Iteration of simulation (-)
X_i	Outputs (-)
D_j^+	Euclidean distance for ideal best (-)
D_j^-	Euclidean distance for ideal worst (-)
v_i^+	Minimum of v_{ij} for non-beneficial criterions; maximum of v_{ij} for beneficial criterions
$x_{ij}^{'}$	The normalized value of criterion *j* for alternative *i* (-)
v_{ij}	Weighted normalized matrix (-)
x_j	The set or vector of all values for criterion j across all alternatives (-)

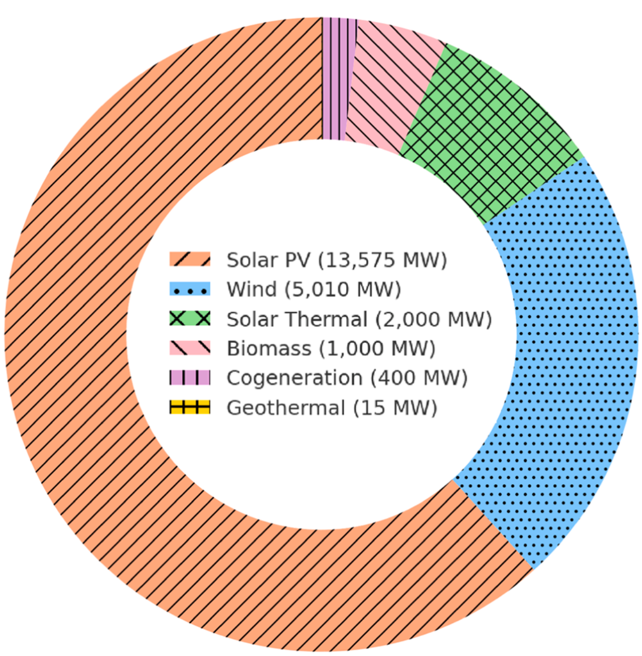


Fig. 1. Analysis of the targeted renewable energy mix in Algeria based on installed capacity by 2030 [7].

planners.

Beyond identifying raw capacity, conducting a sensitivity analysis on simulated data (e.g., HOMER outputs) allows a deeper understanding of energy system behavior [8]. These analyses reveal which parameters, such as radiation, wind speed, initial cost, or population, have the greatest impact on key indicators such as LCOE, production amount, or CO_2 emissions. This information is very valuable for policymakers and energy planners, as it enables them to focus on sensitive parameters, allocate financial resources more effectively, and reduce investment risks. Using MCDM methods such as SAW or TOPSIS enables comparison of different scenarios in solar and wind stations. These methods provide a clear view of system performance across economic, environmental, and social objectives [9,10]. These comparisons help decision-makers to consider a more balanced combination of criteria rather than choosing solely based on a single index (e.g., just cost). Ultimately, such systematic approaches enable more precise design, informed project prioritization, and facilitate an intelligent energy transition in Algeria.

Given the high potential of renewable resources and the urgent need to design energy policies compatible with climatic and economic conditions, identifying optimal solar and wind stations is only the starting point of the analysis journey. What distinguishes this study from previous research is the integration of policy-based MCDM (SAW and TOPSIS) with sensitivity analysis under both deterministic and probabilistic frameworks (Monte Carlo simulation with 5000 iterations. Through this methodological combination, not only the dynamic behavior of the energy system under key parameter changes is revealed, but regional and policy-driven priorities can also be assessed through comparative scenarios. Such a framework provides the possibility to simultaneously evaluate technical, economic, environmental, and social considerations and represents a crucial step towards developing localized analytical tools for smart energy planning in Algeria. In this regard, the review of previous studies in this field below clarifies the position of this research within the existing literature.

Koussa and Koussa in 2015 [11] analyzed the feasibility of grid-connected PV and wind systems at a coastal site in Algeria. Simulation was carried out using MATLAB-SIMULINK, and economic analysis was conducted with HOMER. The results showed that a combination of PV (3 MW) and wind (1.98 MW) with a 24 % share of renewable energy had the lowest LCOE and was the most suitable option, provided that the grid electricity price was $3/kWh.

Bentouba and Bourouis in 2016 [12] assessed the technical and economic feasibility of a hybrid system for providing electricity to the city of Timiaouine in southwestern Algeria. With an average daily solar radiation of 7.82 kWh/m^2 and wind speeds ranging from 5 to 11 m/s, a hybrid solution of wind, PV, and diesel generator was examined. The results showed that this system could supply 100 % of the city's electricity demand with a LCOE of $0.176. Additionally, annual carbon emissions were reduced by 593.125 tons.

Dekkiche and colleagues in 2017 [13] configured a photovoltaic-wind-diesel-battery system to meet a daily demand of 54 kWh in an isolated site with 10 inhabitants in the city of Chlef. Using HOMER, the technical-economic analysis showed that the best option was a photovoltaic-diesel-battery hybrid system, producing 30,472 kWh of energy annually with a LCOE of $0.224. The optimal system included 20 kW of PV panels, a 2.5 kW diesel generator, and 36 batteries with a 4 kWh capacity.

Guezgouz and colleagues in 2021 [14] analyzed, for the first time, the spatiotemporal complementarity between solar and wind resources in Algeria based on the Spearman correlation coefficient at hourly, daily, and monthly scales. The results showed a moderate complementarity in the coastal and southern regions of the country. The highest complementarity was observed in the Annaba province with a coefficient of −0.52. Data showed that electricity production and resource potential had little effect on the complementarity index. The maximum solar and wind potential was reported as 2.38 and 3.33 MWh/m^2 per year, respectively.

Yahiaoui and Tlemçani in 2023 [15] analyzed three hybrid system configurations based on PV, wind, electrolyzer, hydrogen storage, and fuel cell for the city of Timimoun. Using HOMER, the system's performances were evaluated. The PV/Wind/Battery system had a total net present cost (NPC) of $22,621,932 and a LCOE of $1.673/kWh, making it the most expensive option. The second system (PV/Wind/Electrolyzer/H_2 Tank/Fuel Cell) had a total NPC of $14,945,818 and a LCOE of $1.105, also deemed costly. The third option, with a total NPC of $12, 322,474 and a LCOE of $0.912, was identified as the cheapest and most optimal configuration.

Bensalmi and colleagues in 2024 [16] conducted a study aiming to analyze and determine the optimal sizing for a hybrid energy system in six regions of Algeria, including Adrar, Ouargla, Béchar, Sétif, Algiers, and Tiaret. The main objective of this research was to reduce the cost of energy production and enhance the reliability of continuous electricity supply in remote areas. The methodology was based on simulation with HOMER Pro software, modeling a combination of energy sources including photovoltaic panels (PV), wind turbines (WT), and battery storage systems. Climatic data such as solar radiation, ambient temperature, and wind speed were extracted from the NASA POWER database, and the load profile was assumed to have an average daily consumption of 11.26 kWh. Three different configurations were evaluated: PV/WT/Batt, PV/Batt, and WT/Batt. In the PV/WT/Batt hybrid configuration, the city of Adrar achieved the best performance with the lowest LCOE of €0.503/kWh and the lowest percentage of unmet load at 0.009 %, while in Tiaret, the same configuration had the highest LCOE at €0.564/kWh. In the PV/Batt configuration, the LCOE ranged from €0.520 in Béchar to €0.603 in Sétif, with the highest unmet load observed in Tiaret at 0.032 %. The WT/Batt configuration showed the highest costs, with a minimum LCOE of €1.03 in Adrar and a maximum of €1.43 in Tiaret. Also, in this configuration, the percentage of unmet load reached 0.062 % in Ouargla, the highest among the regions. The results of this study showed that the use of the PV/WT/Batt hybrid system had superior performance compared to other configurations, especially in regions with strong renewable resources such as Adrar. Furthermore, the study confirmed the high capability of the HOMER Pro tool in the technical and economic analysis of renewable energy systems across different climates of Algeria.

Dernouni and colleagues in 2024 [17] conducted simulations for a 100 kW Dish Stirling system in five southern Algerian regions, including Adrar (Bordj Badji Mokhtar), Illizi (Djanet), Tamanrasset (Ain Mertoutek), Tindouf, and Béchar, to compare it with a photovoltaic technology. The results showed that the Dish Stirling technology outperformed the 100 kW photovoltaic system in all these regions, demonstrating the highest potential in the Sahara Desert area. This system produces an average of 256 MWh of electricity annually and simultaneously reduces CO_2 emissions by 177 tons. Among the regions studied, Illizi (Djanet) exhibited the best performance, where the Dish Stirling system was able to produce 288.43 MWh of energy annually, enough to meet the annual needs of 230 households. Also, the final LCOE for this system was estimated at only $0.0378 per kWh. Comparisons with previous studies show that the low cost and high efficiency of this technology in southern Algeria—mainly due to high direct normal irradiance—make it a suitable option for clean energy in border and remote areas.

Bouroumeid and colleagues in 2024 [18] conducted a technical and economic study on the integration of photovoltaic systems into building electricity networks in Algeria. The study site was an unused parking lot adjacent to the Faculty of Electrical Engineering at the University of Sidi Bel Abbès, located in northwest Algeria. In this research, initially, using the PVSOL software, a precise 3D model of the parking roof was constructed to evaluate the maximum capacity for installing solar panels, considering factors such as shading, dimensions, orientation, and slope. Then, technical and economic analysis of two scenarios was carried out using the HOMER Pro software: one with electricity supply solely from the conventional grid, and the other with a combination of grid electricity and the installed photovoltaic system. The modeling results showed that 408 Trina Solar 245TSM photovoltaic panels with a nominal power of 245 W could be installed at this location. The system also included six SMA15 DC-AC inverters. In the daily performance analysis, the system's maximum output reached about 90 kW at noon. The 25-year economic analysis showed that although the hybrid scenario (grid - PV) initially required a higher investment (around €50,000), after about 10 years, costs decreased and this scenario became more profitable than the grid-only case. Also, only one inverter replacement in the fifth year and annual maintenance costs were considered.

Touahri and colleagues in 2024 [19] used HOMER software to conduct a technical-economic analysis of hybrid PV/Diesel systems for supplying electricity to isolated households in the hot and desert climate of ADRAR. The input data included hourly solar radiation and an annual load equal to 11.2 kWh/day. The results showed that the diesel system, with a LCOE of $0.407/kWh and a fuel price of $0.14/liter, was an expensive and polluting option. Among the configurations studied, a combination of 2 kW of PV panels and a 2.3 kW diesel generator was economically more affordable, with a LCOE of $0.172/kWh. Moreover, annual CO_2 emissions were reduced by about 5 tons compared to the standalone diesel system.

Haddouche and colleagues in 2024 [20] also conducted a similar study in the Timimoun region of Algeria. They analyzed a hybrid wind/diesel/battery system for a village with about thirty households. Wind speed (ranging from 3.99 to 5.42 m/s) and daily load (0.9 to 3.6 kWh) were analyzed using HOMER software. The best configuration included a WES 5 Tulipo turbine with a nominal power of 2.6 kW and a diesel price of $0.22/kWh. This system reduced the LCOE to $0.569 and increased the share of renewable energy to 86 %. CO_2, CO, and SO_2 emissions were also reduced by >70 %. Sensitivity analysis introduced this system as the optimal solution at wind speeds above 3.5 m/s.

Mathaba et al. (2024) [21] conducted a study in Mahaneng village, South Africa, developing a novel MCDM framework combining multi-objective optimization with metaheuristic algorithms and VIKOR ranking for the optimal design of a hybrid renewable energy system. The study followed three stages: nonlinear programming-based modeling, multi-objective optimization using a LaF/PSO hybrid algorithm, and evaluation of alternatives via MEREC-VIKOR. Optimization targeted minimizing LCOE, CO_2 emissions, and renewable energy waste while ensuring high reliability. Simulation results for a residential load (17.4 kW peak, 191 kWh/day average) identified a top-performing configuration with $0.199/kWh LCOE, zero energy waste, and 982 tons of CO_2 emissions over a 25-year operational period.

Karbassi Yazdi et al. (2024) [22] employed a hybrid MCDA approach (Delphi, MEREC, Fermatean fuzzy WASPAS) to identify optimal sites for green energy projects across India. Sites such as NP Kunta and Bhadla were evaluated under uncertainty. Political and strategic factors were found to be the most influential, with NP Kunta scoring highest (Qi = 1.4719).

Magableh et al. (2025) [23] explored sustainable green energy strategies in the UAE using innovative fuzzy MCDM methods. Data were collected through expert surveys, official statistics, and stakeholder consultation. A team of ten experienced experts—including senior decision-makers, academics, and consultants—evaluated strategies using a hybrid framework incorporating Fuzzy MOORA, Fuzzy SAW, and combinations such as FMOORA-FSAW and MOORA-SAW. Results showed that the MOORA-SAW hybrid aligned best with expert opinions, favoring "open" and "integrated" strategies under equally weighted criteria.

Dash et al. (2025) [24] assessed sustainable wind energy resources for rural regions in India and Spain using TOPSIS and PAPRIKA. Seven wind energy system types—including onshore, offshore, floating, vertical-axis, micro, hybrid, and storage-backed farms—were evaluated against criteria such as capacity factor, output, LCOE, wind assessment, turbine performance, environmental impact, grid integration, policy frameworks, and community participation. Onshore turbines scored highest (69.9), followed by vertical-axis (66.5) and hybrid systems (60.8), while others ranked lower due to high cost and integration challenges.

Gaurav et al. (2025) [25] designed an off-grid hybrid energy system for a local healthcare center on Andrott Island, India, using HOMER Pro for simulation and MCDM (TOPSIS, VIKOR) in MATLAB for ranking. Ten configurations combining solar PV, wind, biomass, diesel, and batteries were assessed. The H7 configuration (PV-wind-battery) was optimal for environmental performance, while H10 (PV-wind-biomass-battery) offered the lowest LCOE ($0.1335/kWh) and minimal CO_2 emissions (2.5 kg/year).

Goswami et al. (2025) [26] applied a hybrid MCDM framework (CRITIC, EDAS, CODAS, CoCoSo) to rank renewable sources for power plant development in India. Based on six competitive technical, economic, and environmental criteria, hydropower emerged as the top choice, followed by solar. Geothermal and biomass were ranked lowest. Sensitivity analysis revealed that EDAS and CoCoSo outperformed CODAS in ranking precision.

While many previous studies have employed MCDM approaches without sensitivity analysis, the present work integrates two decision-making techniques (TOPSIS and SAW) with Monte Carlo simulation, offering a more robust assessment of decision stability under real-world uncertainty. This study focuses on applying MCDM to hybrid renewable energy systems, addressing an important research gap. Most previous works either performed only techno-economic analyses with HOMER or applied MCDM without integrating policy-oriented scenarios and uncertainty modeling. In this study, for the first time, the two decision-making methods SAW and TOPSIS are employed simultaneously, along with Monte Carlo sensitivity analysis to evaluate the stability of rankings under varying criterion weights. The methodological innovation of the paper is structured around three main pillars:

- The simultaneous use of seven diverse evaluation criteria, including initial cost, LCOE, payback period, renewable energy production, energy losses, CO_2 emissions, and population.
- The definition of six policy-oriented scenarios to assess priorities under various decision-making perspectives;
- Statistical validation of result stability through 5000 iterations of Monte Carlo analysis.

Additionally, the present study is conducted in two climatically and demographically distinct regions in Algeria (Tamanrasset and Adrar), and the optimal resource configurations for each region are presented individually. Unlike most previous studies, which have focused on a limited number of configurations or criteria, this research framework—through scenario analysis, uncertainty assessment, and combined use of decision-making methods—offers a deeper understanding of decision-making robustness under shifting preferences. Therefore, beyond its methodological novelty, this study provides an effective decision-support tool for policymakers, regional planners, and development agencies in selecting optimal systems aligned with economic, social, or environmental objectives. Although only two representative locations (Tamanrasset and Adrar) were selected to reflect highly contrasting climatic conditions, the decision space was expanded through six scenario-based weighting schemes. These scenarios enabled a deeper comparative analysis, effectively transforming the evaluation into a multi-layered MCDM exercise beyond a simple two-site comparison.

2. Study locations

Algeria, as one of the largest countries in Africa, is located in the northern part of the continent, and a large portion of it lies within the Sahara Desert region [27]. This geographical location provides unique climatic conditions for the development of renewable energies [28]. In terms of solar resources, Algeria possesses one of the highest annual solar irradiance levels in the world; in fact, the average daily solar radiation in the southern regions of the country exceeds 7.5 kWh/m^2-day [29]. Additionally, the western and southwestern regions of the country have significant wind potential, with average wind speeds in some areas exceeding 5 m/s, which is quite suitable for wind energy exploitation [30].

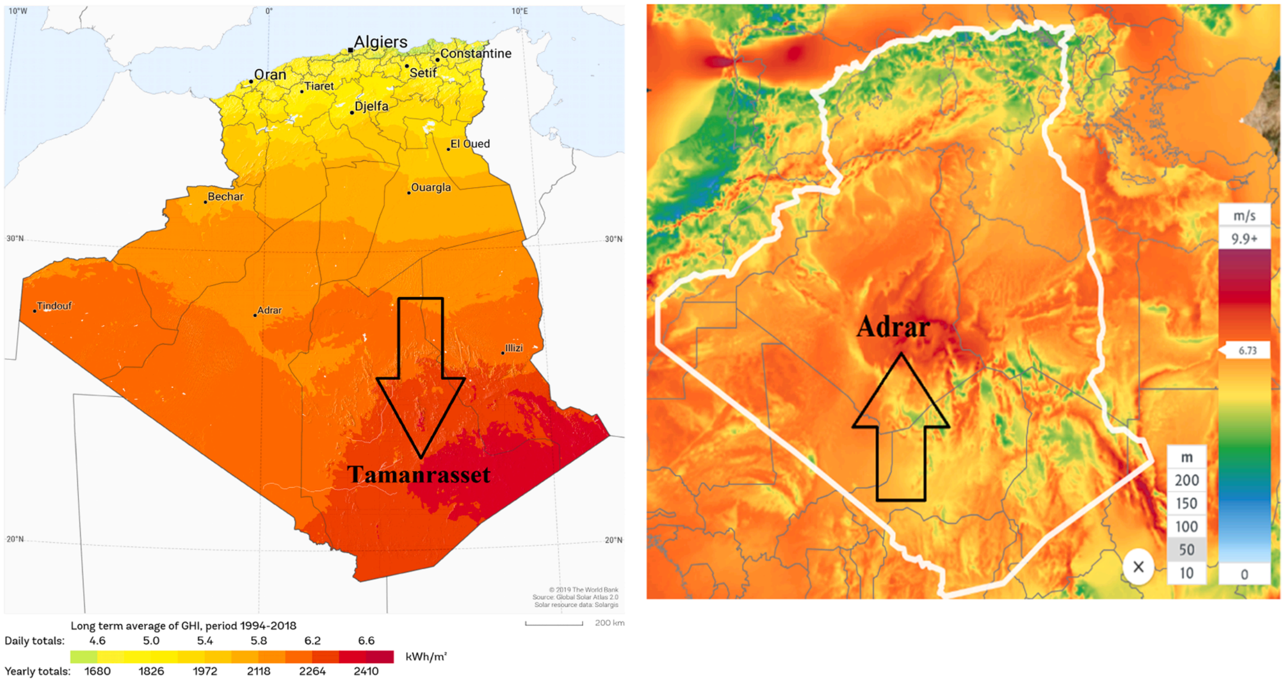
In this study, two stations, Tamanrasset and Adrar, were purposefully selected to examine two contrasting yet complementary scenarios of the country's renewable energy potentials:

- Tamanrasset, located in southeastern Algeria deep within the Sahara Desert, has the highest level of solar irradiance in the country. This station represents extremely sunny regions with minimal light pollution and atmospheric dust, making it an ideal option for the development of solar systems.
- Adrar, located in southwestern Algeria, is one of the windiest areas in the country. This region represents wind-rich areas that have high potential for wind energy exploitation.

Selecting these two locations captures the full spectrum of Algeria's renewable climatic zones. It also enables precise analysis of how different climatic conditions affect the design and optimization of hybrid energy systems. In Figs. 2(a) and 2(b), the geographical locations of Tamanrasset and Adrar are shown, respectively, on the solar irradiance map [31] and the wind speed map [32] of Algeria. Fig. 2 illustrates the distinct differences in energy resource patterns between these two regions, reinforcing the rationale behind their purposeful selection.

3. Methodology

The methodology of this study is organized in a stepwise framework to evaluate hybrid renewable energy systems in Algeria. First, two representative stations—Tamanrasset (solar-based) and Adrar (wind-based)—were selected as case studies. Second, HOMER software was employed to simulate and optimize hybrid energy configurations, providing key outputs such as cost, renewable electricity generation,



a) b)

Fig. 2. Geographical location of the selected stations on: (a) solar irradiance map, (b) wind speed map.

losses, excess energy, and CO_2 emissions. Third, these outputs, together with a social indicator (population coverage), were defined as the evaluation criteria. Fourth, two widely used MCDM methods, SAW and TOPSIS, were applied under six different weighting scenarios to rank the alternatives. Finally, both deterministic and Monte Carlo-based sensitivity analyses were conducted to test the robustness of the obtained rankings. This systematic framework ensures that the methodology is comprehensive, transparent, and aligned with the study's policy-oriented objectives.

Fig. 3 presents a graphical summary of the research process conducted for the technical, economic, and environmental assessment of two selected renewable energy stations in Algeria. Two key sites were identified: the solar station in Tamanrasset (southern Algeria, with very high irradiance) and the wind station in Adrar (western Algeria, with favorable wind speeds). These were selected as the country's best regions in terms of renewable energy resources.

In the first step, regional climatic data, including solar irradiance and wind speed, were entered into HOMER v2.81 software. Through simulation, the optimal configuration of systems at each site was extracted [33]. In both cases, a diesel generator was considered as a backup to ensure a stable electricity supply for a residential unit.

Subsequently, advanced statistical analyses such as uncertainty

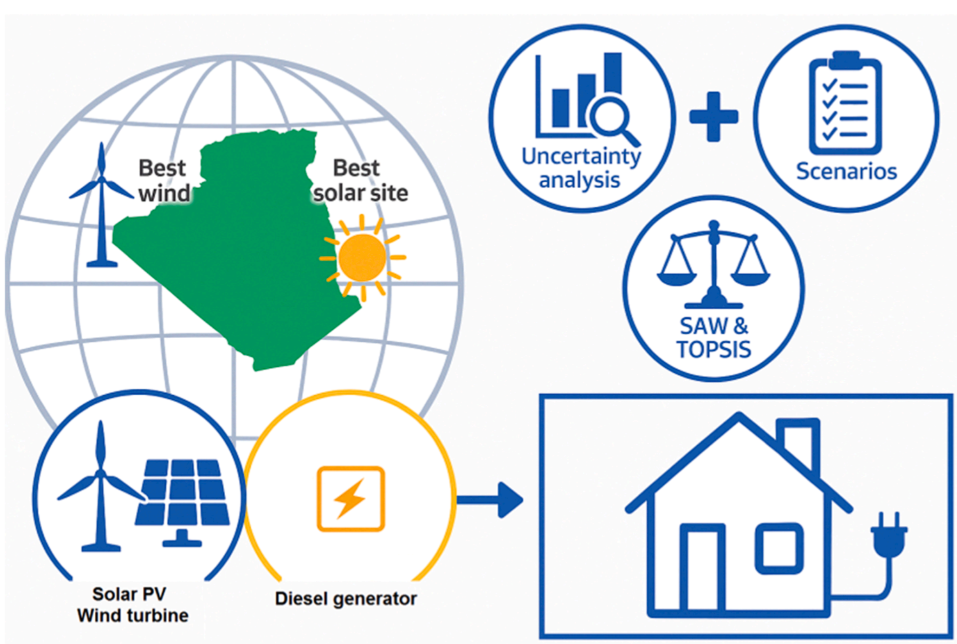


Fig. 3. Process of evaluating solar and wind energy stations in Algeria using a multi-criteria analysis and energy simulation approach.

analysis, parameter sensitivity analysis, and comparison of outputs under different scenarios were defined [34]. Each scenario assigns different weights to criteria such as LCOE, CO_2 emissions, renewable energy production, energy losses, population coverage, and others. The output of this stage is a scoring matrix for each station under each scenario.

In the final stage, to compare the options, two MCDM methods [35], namely SAW and TOPSIS, were employed [36]. The ranking results of these two methods, presented in a comparative manner, illustrate the relative advantage of each station across different scenarios. This multi-criteria comparison aims to provide intelligent strategies for prioritizing clean energy projects and delivering practical data to decision-makers in the energy sector.

Overall, Fig. 3 illustrates the hierarchical sequence of the research steps from raw climatic data to final decision-making analysis, and demonstrates the logical connection between simulation, statistical analysis, multi-criteria modeling, and spatial evaluation stages. This graphical framework can serve as a model for similar regional analyses in other countries with different climatic and policy conditions [37]. The methodological novelty of the present work lies in coupling simulation outputs from HOMER with scenario-driven MCDM and embedding both gradual and probabilistic sensitivity analyses.

3.1. Rationale for selecting SAW and TOPSIS

In the present work, among the various MCDM methods, SAW and TOPSIS were selected, based on several technical considerations. First, both methods are computationally simple and easy to understand, making them highly suitable for energy policy-related decision-making. Second, their capacity to analyze diverse scenarios involving economic, technical, environmental, and social criteria with varying weights makes them well-suited for the multi-criteria structure of this research. Third, these methods integrate effectively with statistical techniques such as Monte Carlo simulation, allowing for the examination of decision stability under weight uncertainty. Moreover, the simultaneous use of both methods aims to assess consistency and robustness in the rankings, whereby comparison of the results serves as a form of cross-validation. Therefore, the combination of SAW and TOPSIS not only enables a multidimensional analysis of the problem but also ensures precise and reproducible implementation of the model.

3.2. Rationale for selecting HOMER software

HOMER software was chosen as the primary tool for simulation and optimization, given its accuracy and strong reputation in modeling hybrid energy systems for remote or resource-limited regions. The software utilizes hourly data over the project lifetime to calculate key technical, economic, and environmental indicators such as energy production, losses, LCOE, payback period, and CO_2 emissions. These outputs precisely correspond to the seven criteria defined in the MCDM framework. Furthermore, HOMER determines the optimal system configurations based on defined constraints (load profile, fuel prices, climatic resources, etc.), thereby generating realistic and reliable data for multi-criteria analysis. In the absence of such detailed simulation, MCDM analysis would inevitably rely on hypothetical or generalized data, which would undermine the credibility of the results. Therefore, HOMER plays a fundamental role in this study, bridging the gap between actual system performance and structured decision analysis.

Compared to metaheuristic optimization approaches such as Genetic Algorithms or Particle Swarm Optimization, HOMER offers an integrated environment where technical, economic, and environmental indicators are simultaneously modeled using real resource data. While metaheuristic methods are powerful for design optimization, they typically require additional coding and assumptions, and their outputs may lack transparency for policy-level decision-making. Hence, HOMER was selected as a more practical and widely validated tool for this study.

4. Governing equations

In the present work, two multi-criteria decision-making methods—SAW and TOPSIS—were applied simultaneously to analyze and rank renewable energy policy scenarios. These methods were selected due to their computational simplicity, high discrimination ability, and compatibility with statistical analysis. Their joint application allows cross-validation of ranking stability. To assess sensitivity to variations in criterion weights, Monte Carlo analysis with 5000 iterations was conducted, significantly enhancing the robustness and reliability of results. All computational steps—including data normalization, weight assignment, final score extraction, and sensitivity analysis—are presented step-by-step with mathematical formulations to ensure full reproducibility of the study.

In the design and simulation of hybrid renewable energy systems, the extraction and analysis of the governing equations for the performance of various system components play a fundamental role in understanding and optimizing the final structure. In Fig. 4, the physical relations related to power production by PVs and WTs, battery behavior, inverter efficiency, as well as the calculation of system economic indices, are explicitly presented. These fundamental equations play a key role in the computations performed by the HOMER software and are reliable in economic, environmental, and technical analyses.

The output power of the solar system depends on solar irradiance intensity, panel efficiency, and environmental conditions [38,39]. The solar module output depends on instantaneous solar irradiance ($\overline{H_T}$), extraterrestrial irradiance ($\overline{H_{T.STC}}$), the panel's rated capacity (Y_{PV}), and the incident irradiance factor (f_{PV}). It is calculated as the direct product of these components. Regarding WTs, the output power equation is defined based on fluid mechanics principles and the law of conservation of energy [40]. Wind power is a function of air density (ρ), rotor swept area (A), the cube of wind speed (v^3), and the turbine power coefficient (C_p), which typically ranges from 0.25 to 0.45. On the other hand, the performance of batteries [41], which are used to store energy during periods of renewable resource shortage, is defined by parameters such as capacity under different states (P) and overall efficiency (η). The efficiency of DC/AC converters [42] is also another factor contributing to the overall system efficiency and is generally considered as a percentage factor in calculations. In economic calculations, the total NPC index plays an important role in the economic evaluation of the system [43]. This index, by considering annual costs ($C_{ann,\ total}$) and applying the interest rate (i), calculates the total discounted cost over the system's lifetime. Based on this value, the LCOE [44] is calculated, which serves as the primary cost-comparison metric for systems in comparative analyses. This index is obtained by dividing the total NPC by the total energy generated over the system's lifetime and is expressed in $/kWh.

The HOMER simulations provide outputs for multi-criteria analyses (SAW and TOPSIS), including initial cost, LCOE, renewable electricity production, system losses, excess energy, and CO_2 emissions, along with the regional population index. The analysis of these parameters within different weighting scenarios helps decision-makers to identify optimal options by considering local conditions and policymaking requirements.

In the present work, in order to compare criteria with different scales, linear Min-Max normalization was employed. This method transforms the original values into the [0,1] range, enabling direct comparison among the criteria. The mathematical formulation of this type of normalization is as follows [45]:

For benefit criteria:

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \tag{1}$$

For cost criteria:

$$x'_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \tag{2}$$

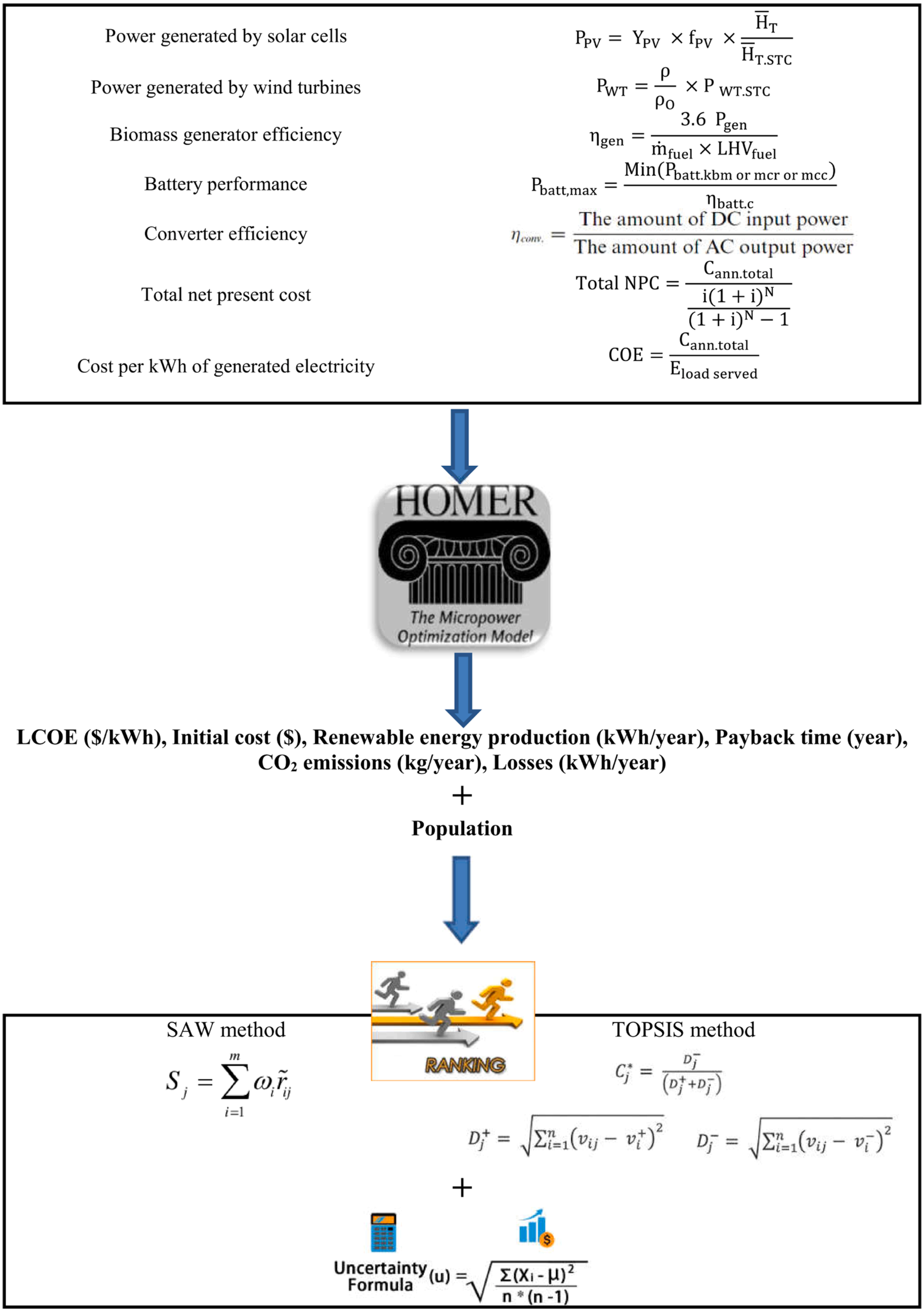


Fig. 4. Governing equations of the problem.

In these equations, x_j represents the set (or vector) of all values for criterion j across all alternatives, x_{ij} represents the value of criterion j for alternative i, and x'_{ij} denotes its corresponding normalized value.

5. Input data

The schematic of the studied system is presented in Fig. 5. A battery and a diesel generator are used as backup systems under emergency conditions, and the goal is to supply electricity for a residential apartment. In Table 1, the input data required for the simulation, including technical and cost information of the utilized equipment, are provided.

Additional input data required for the simulation are as follows:

The price of diesel fuel is $0.217 per liter [49], the project lifetime is 25 years [50], and the real annual interest rate is 3 % [51]. The penalty for emissions is considered zero per ton [52] in the simulations. Currently, in Algeria, there is no enacted legislation or formal policy in place that imposes financial penalties on greenhouse gas or other pollutant emissions by the government. In other words, unlike certain countries that have implemented measures such as a carbon tax, no such mechanism has yet been established in Algeria. Had a non-realistic value for the penalty been applied in the modeling, the resulting outcomes of the study would not have aligned with the actual policymaking and decision-making context of the country, thereby undermining its practical applicability for local policymakers. Therefore, in order to preserve the accuracy and relevance of the results to Algeria's legal and operational realities, this parameter was set to zero in the model.

In Fig. 6, the most important simulation input—namely, the electricity consumption profile that needs to be supplied—is shown. The average annual electricity demand is 7.81 kWh/day, and to account for random variations, a 15 % daily randomness and 20 % hourly randomness have been applied. The daily electricity consumption value of 7.81 kWh was selected based on the average consumption of urban apartment units in Algeria, which aligns with the official annual consumption data (approximately 2400 kWh/year) [53].

Fig. 7 presents a combination of the monthly averages of three key climatic indicators: daily solar radiation, clearness index, and wind speed. Solar radiation data indicate that the highest solar energy is received during the period from May to August, with values approaching 8 kWh/m^2/day, which reflects an excellent potential for solar energy utilization in this timeframe. The clearness index, which represents the ratio of surface-level solar radiation to the top-of-atmosphere radiation, remains in a relatively stable range of 0.72 to 0.76 during most months. This reflects a relatively clear atmosphere and suitable conditions for the stable performance of photovoltaic systems. On the other hand, wind speed fluctuates between 6 and 6.9 m/s throughout the year, reaching its peak during the spring season. The simultaneous analysis of these three indicators allows for a comprehensive evaluation of the combined renewable energy potential in a selected region. This is particularly effective for the design of hybrid solar-wind systems, as the relative overlap between high solar radiation and favorable wind speed in certain months can lead to optimized annual performance and increased capacity factor.

The heat map presented in Fig. 8 illustrates the distribution of weight assignments to the seven decision-making criteria across six different MCDM scenarios. These scenarios are designed with various approaches to address the diverse priorities of policymakers, engineers, and energy planners. The first scenario (Cost-effective) heavily emphasizes initial cost (30 %), electricity generation cost (25 %), and payback period (25 %), making it highly economically-oriented and suitable for projects with tight budget constraints. The second scenario (Environment-oriented) focuses particularly on reducing CO_2 emissions (35 %) and minimizing energy losses (20 %), making it appropriate for projects driven by sustainability and environmental objectives. The third scenario (Performance-oriented) assigns high weights to renewable energy production (35 %) and losses (20 %), targeting systems where technical efficiency and energy performance are prioritized. The fourth scenario (Balanced) considers relatively equal weights for all criteria and remains unbiased, thus being suitable for impartial evaluation of options. The fifth scenario (Policy-driven) combines the goals of clean energy production, pollution reduction, and population coverage, and is highly effective for large-scale, national-level energy transition planning. Finally, the sixth scenario (Human-centered) assigns the highest weight to population (35 %) and a significant share to renewable energy production (25 %), reflecting an approach based on delivering effective services to the largest number of residents and maximizing social impact. This heat map reveals the differences in goals and focus areas across the scenarios and provides a transparent tool for selecting the optimal option under varying conditions.

These weights were determined through a combination of literature review, expert consultation, and logical alignment with policy priorities. For instance, in the cost-oriented scenario, the emphasis on initial cost (30 %), electricity generation cost (25 %), and payback period (25 %) is based on numerous studies and energy investment guidelines in developing countries, which identify these factors as primary constraints. Likewise, the 35 % weight assigned to CO_2 emissions in the environmental scenario is grounded in global carbon reduction criteria and prevailing policy trends in the field of sustainability. In the human-centered scenario, the 35 % weight allocated to the population criterion is based on the principle of expanding service coverage and

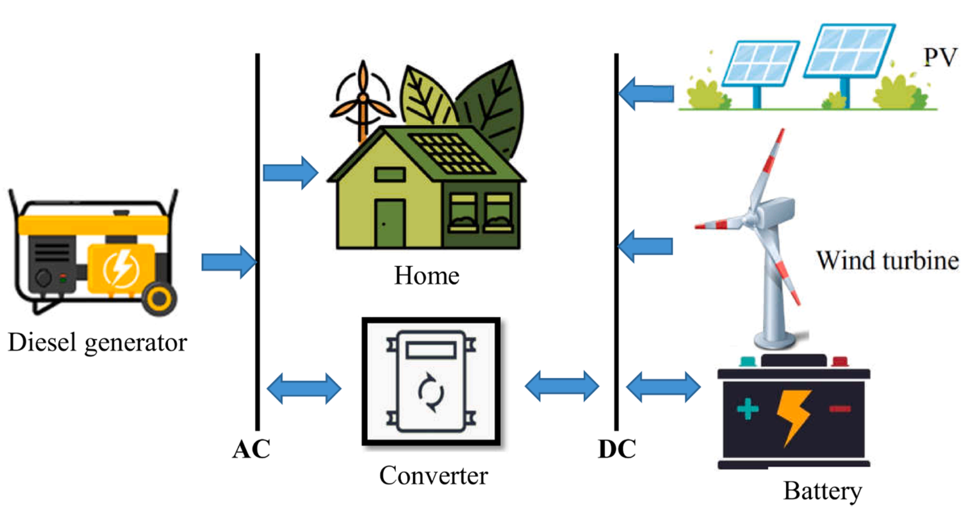


Fig. 5. Schematic of the system studied in the present work.

Table 1
Input data required for simulation using HOMER.

Equipment	Cost ($)			Size (kW)	Other technical information
	Capital	Replacement	Operating & Maintenance		
PV [46]	350	350	10	0–10	Lifetime: 25 years, Derating factor: 90 %
WT [47]	850	850	10	0–10	Lifetime: 20 years, Hub height: 17 m Type: Genreic 1 kW
Battery [46]	174	174	5	0–10	Type: Trojan T-105, Lifetime: 845 kWh
Converter [46]	138	138	10	0–10	Lifetime: 15 years, Efficiency: 95 %
Diesel generator [48]	200	200	0.5	0–10	Lifetime: 15,000 h, Efficiency: 50 %

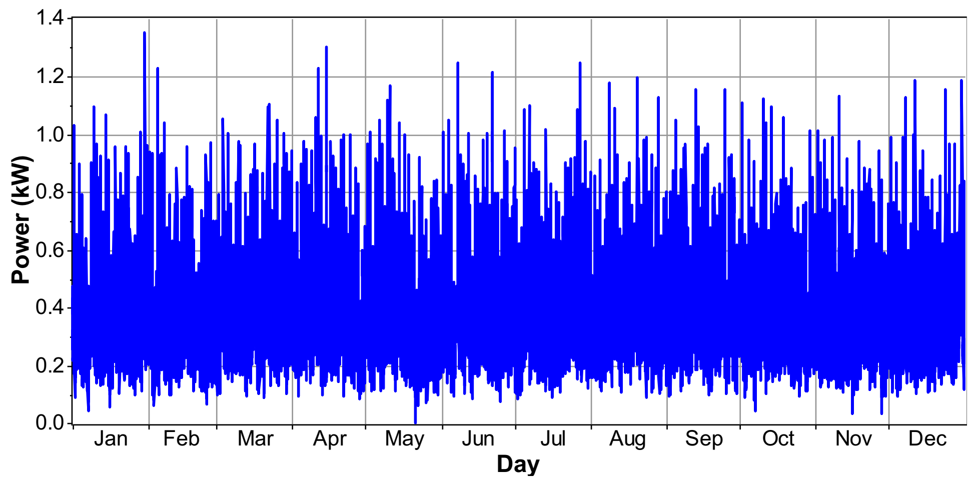


Fig. 6. Electricity consumption profile throughout the year.

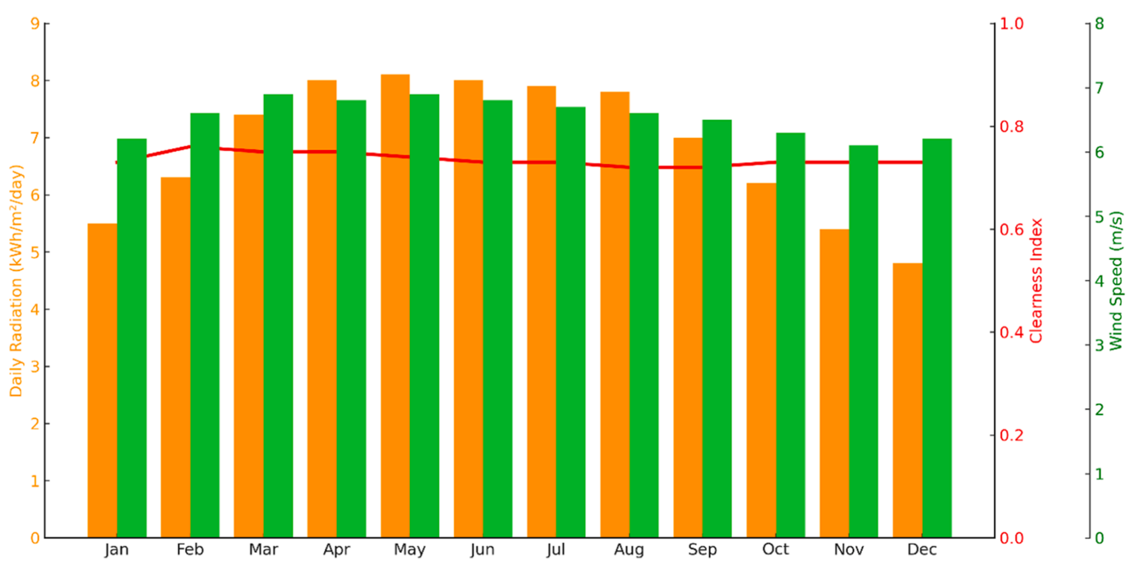


Fig. 7. Monthly average of solar radiation (Adrar station), clearness index (Adrar station), and wind speed (Tamanrasset station).

promoting social equity, as addressed in regional development frameworks. Although these weightings were not derived through a formal pairwise comparison matrix, they were selected based on purposeful, practical prioritization aligned with the actual planning context in Algeria.

6. Results

6.1. Simulation outputs (HOMER results)

Table 2 shows the HOMER simulation results for two Algerian stations, Tamanrasset and Adrar, combining economic, technical, environmental, and demographic indicators for performance comparison.

In Tamanrasset, the optimized system includes 2 kW solar panels and 12 batteries, with no fossil fuel generator. It covers 73,128 people, the highest among the two regions. The initial cost is $3064, with LCOE

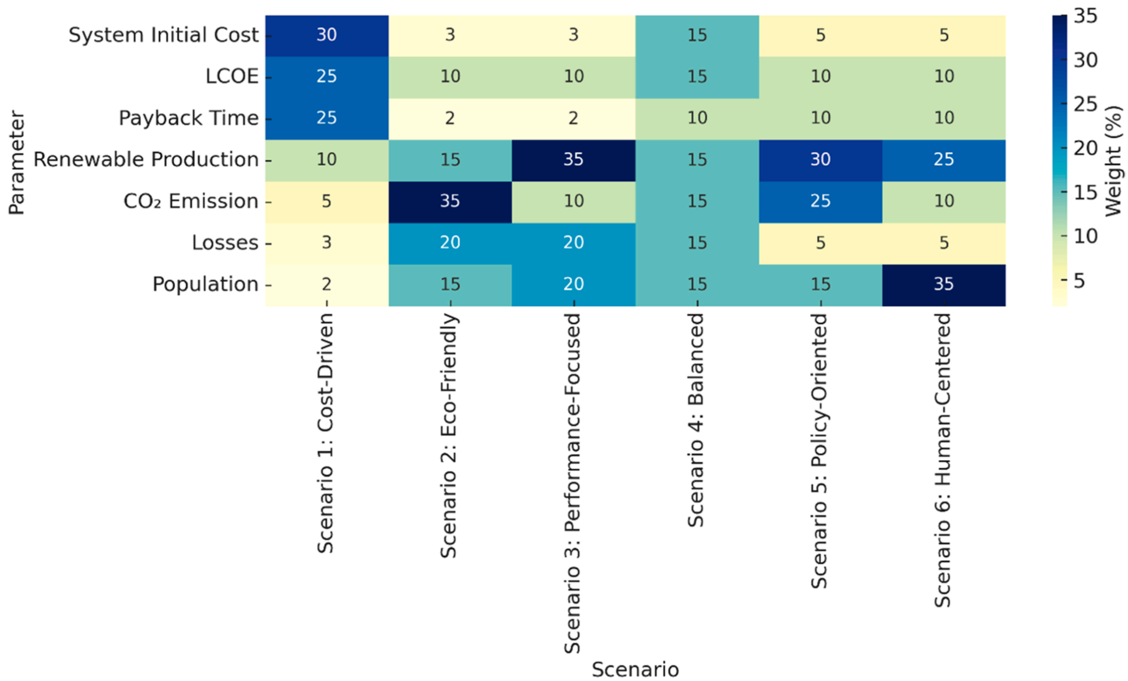


Fig. 8. Weighting scenarios for MCDM.

Table 2
Comparison of key performance metrics for two hybrid renewable energy systems in Tamanrasset and Adrar.

Location	Initial cost ($)	LCOE ($/kWh)	Renewable production (kWh/year)	Losses (kWh/year)	CO_2 emission (kg/year)	Payback time (year)	Population
Tamanrasset	3064	0.225	4619	480	0	0.29	73,128
Adrar	5636	0.257	7425	337	86	0.582	43,903

$0.225/kWh and a very short payback period of 0.29 years. The system produces 4619 kWh/year of renewable electricity, with 480 kWh/year losses but zero CO_2 emissions, making it economical, fast-returning, and fully clean.

In Adrar, the system comprises 3 WTs, 1 kW diesel generator, and 15 batteries, covering 43,903 people. It generates 7425 kWh/year, about 60 % more than Tamanrasset, with losses of 337 kWh/year. However, costs are higher ($5636 initial, LCOE $0.257/kWh) and the payback period is 0.582 years, nearly double Tamanrasset. It also emits 86 kg CO_2/year due to diesel use, though still within an acceptable range.

Overall, Tamanrasset, with lower costs, zero emissions, and faster payback, is ideal for cost- and environment-oriented scenarios, especially in populous regions. Adrar, despite higher costs and emissions, provides greater energy output and efficiency, making it more suitable for performance-focused scenarios or areas with strong wind resources. The inclusion of population ensures assessment of energy justice and

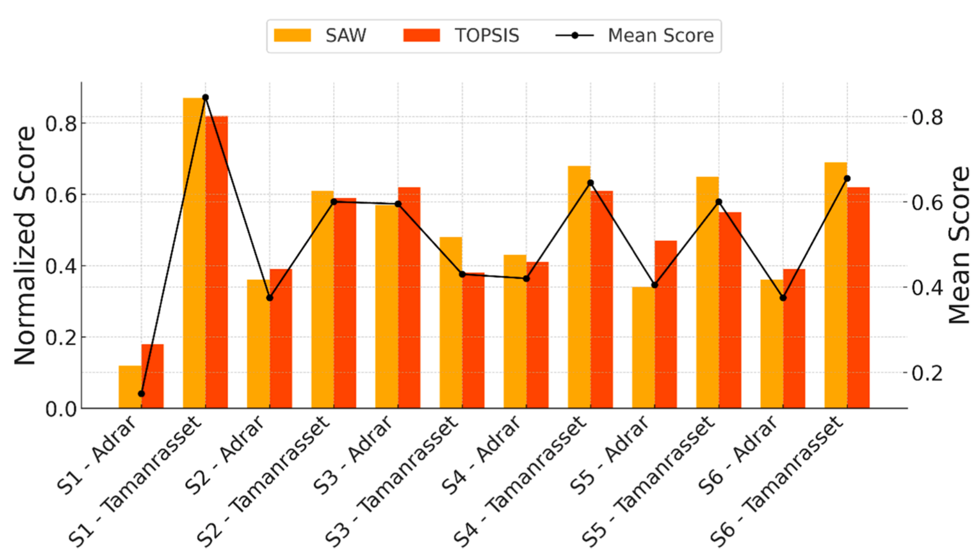


Fig. 9. Normalized SAW and TOPSIS scores for six policy scenarios in Adrar and Tamanrasset, with mean value overlay.

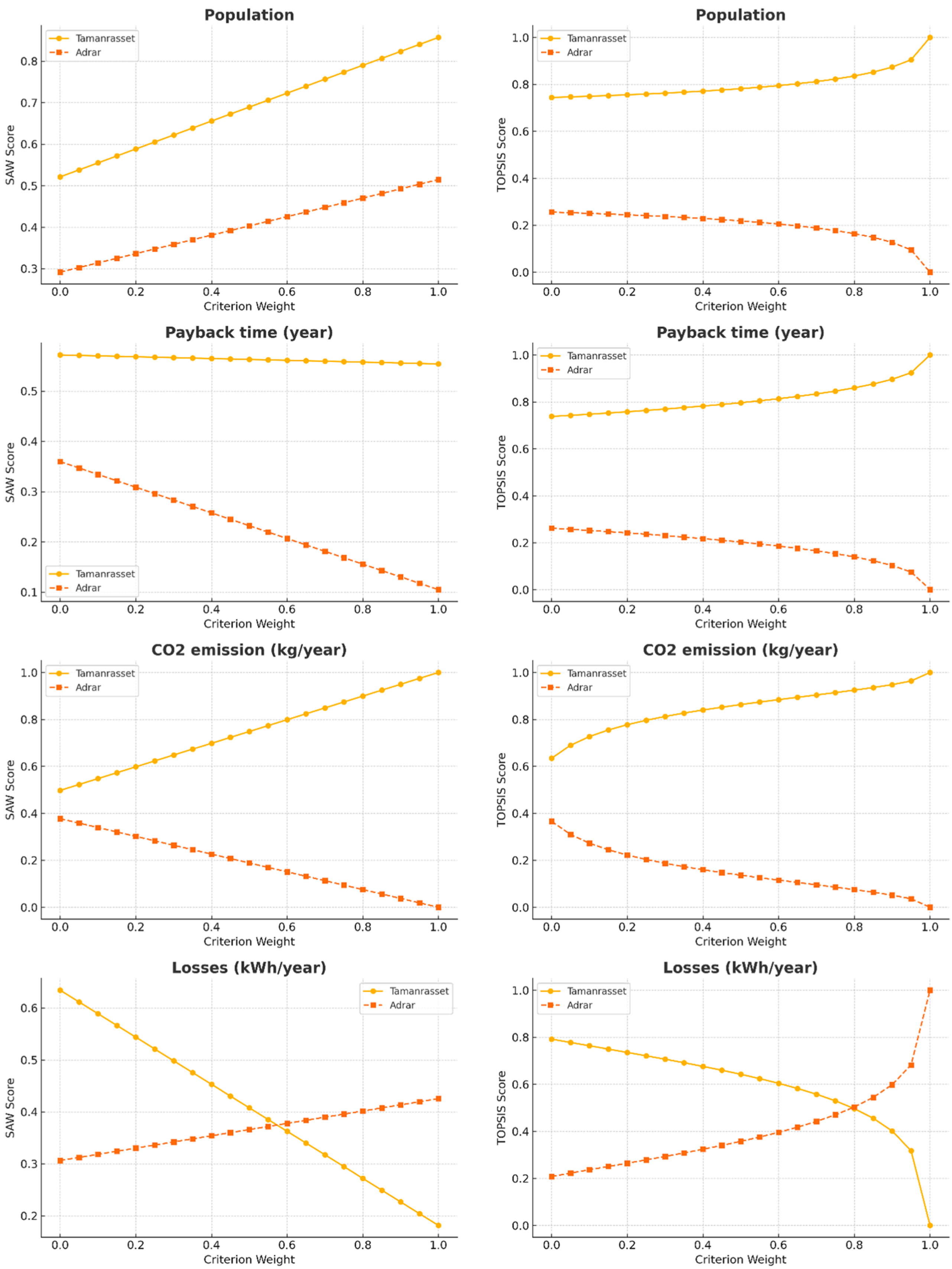


Fig. 10. Sensitivity analysis of hybrid renewable energy system evaluation based on criterion weights using SAW and TOPSIS methods.

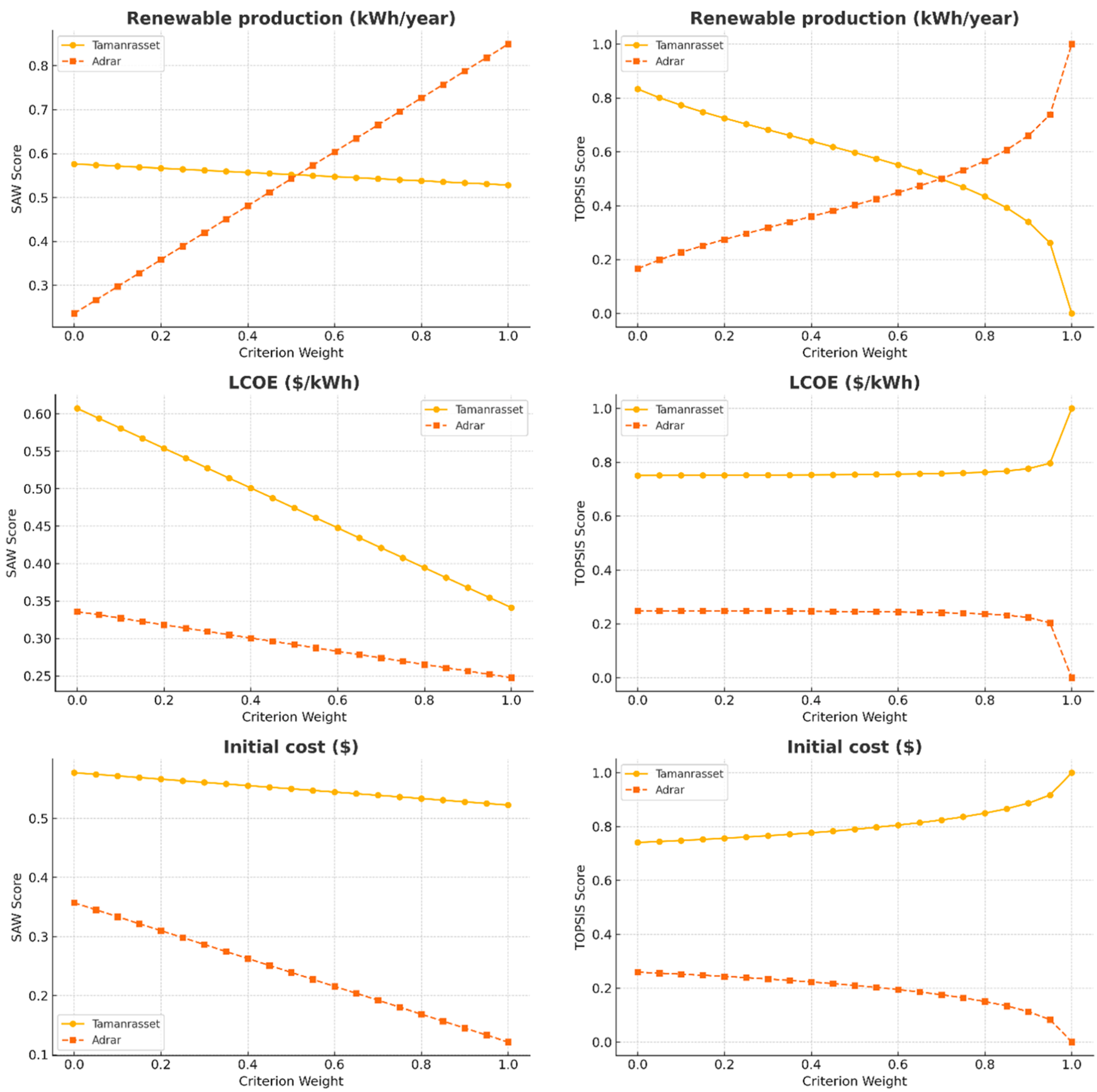


Fig. 10. (*continued*).

regional prioritization.

6.2. SAW and TOPSIS rankings

Fig. 9 shows the final MCDM results using SAW and TOPSIS for ranking two hybrid renewable systems (Tamanrasset and Adrar) under six weighting scenarios, based on seven criteria: Initial Cost, LCOE, Payback Period, Renewable Energy Production, Losses, CO_2 Emissions, and Population.

In the Cost-Driven scenario, focused on economic criteria, Tamanrasset ranked first (SAW: 0.87, TOPSIS: 0.8172), while Adrar was second with much lower scores (SAW: 0.13, TOPSIS: 0.1828).

In the Eco-Friendly scenario, emphasizing CO_2 emissions and losses, Tamanrasset again led (SAW: 0.65, TOPSIS: 0.6126) due to zero emissions, compared to Adrar (SAW: 0.35, TOPSIS: 0.3874).

In the Performance-Focused scenario, highlighting renewable production, Adrar ranked first (SAW: 0.55, TOPSIS: 0.6195), consistent with its higher generation capacity.

With Balanced weights, Tamanrasset again prevailed (SAW: 0.7, TOPSIS: 0.5985), reflecting advantages in cost and emissions.

In the Policy-Oriented scenario, focused on population coverage and clean energy, Tamanrasset remained first (avg. score 0.5844).

Finally, in the Human-Centered scenario, where population had the highest weight, Tamanrasset decisively ranked first (SAW: 0.7, TOPSIS: 0.607), benefiting from nearly twice Adrar's population and its ability to supply clean, low-cost energy.

Overall, Tamanrasset ranked first in five of six scenarios, while Adrar excelled only in the performance-focused case. This confirms Tamanrasset's superiority in economic, environmental, balanced, and social aspects, whereas Adrar dominates when technical performance is prioritized. The combined use of SAW and TOPSIS improved ranking accuracy and reliability for strategic decision-making.

6.3. Deterministic sensitivity analysis

To comprehensively evaluate the impact of criteria weighting in MCDM, Fig. 10 presents the sensitivity of SAW and TOPSIS scores when the weight of each of the seven criteria is varied from 0 to 1. This process reveals the stability of rankings, critical criteria, and robustness of final solutions.

In the Population criterion, both methods favor Tamanrasset due to its larger population. In SAW, the score gap grows linearly, while in TOPSIS, Tamanrasset rapidly approaches the ideal (1.00) after weight >0.6, and Adrar's score approaches zero, showing stronger divergence.

For Payback Time, Tamanrasset maintains advantage with very low payback. In SAW, its score is nearly constant, while Adrar declines steadily. In TOPSIS, Tamanrasset rises sharply after weight >0.6 to 1.00, while Adrar drops to zero, confirming the decisive role of this criterion.

For CO_2 Emissions, Tamanrasset (0 kg) gains linearly to 1.00 in SAW, while Adrar declines symmetrically. In TOPSIS, Tamanrasset grows from ~0.65 to 1.00 and Adrar falls to zero, showing stronger nonlinearity and sensitivity.

In Energy Losses, Adrar (lower losses) gains score as weight increases. In SAW, scores cross at ~0.5; in TOPSIS, crossover occurs later (~0.7–0.8), with Adrar reaching 1.00 and Tamanrasset declining to zero, indicating this criterion can change rankings if heavily weighted.

For Renewable Energy Production, Adrar surpasses Tamanrasset only when the weight exceeds 0.55 in SAW and ~0.65 in TOPSIS, highlighting that its technical advantage affects rankings only under dominant weighting.

For LCOE, both options decline in SAW, but Tamanrasset decreases faster though still ahead overall. In TOPSIS, both remain stable until very high weights, where Tamanrasset reaches 1.00 and Adrar drops to zero.

For Initial Cost, both options decline with weight, but Adrar more steeply. Tamanrasset remains superior across both methods; no rank switching occurs, showing this is a stabilizing rather than decisive criterion.

Overall, SAW exhibits linear and moderate sensitivity with rank switches at mid weights (0.4–0.6), while TOPSIS shows nonlinear, polarized behavior with divergence at high weights. Quantitatively, Tamanrasset outperforms in 5–6 criteria (population, payback, CO_2, cost), while Adrar surpasses only under high weights for 2–3 technical criteria (renewable energy, losses). Practically, Tamanrasset suits socio-environmental priorities, whereas Adrar becomes optimal when technical output dominates. Combined use of SAW and TOPSIS enables robust, priority-aligned decision-making.

6.4. Monte Carlo sensitivity analysis

This study presents, for the first time in this context, a Monte Carlo-based validation of MCDM results using 5000 randomized weighting scenarios for SAW and TOPSIS (Fig. 11). In each iteration, a random weight vector (sum = 1) was generated, applied to normalized values, and the final scores of Tamanrasset and Adrar were calculated. Beyond rankings, the score difference (Tamanrasset − Adrar) was also extracted and plotted as a histogram to capture the full probabilistic behavior of the model. This enabled accurate statistical determination of how often one option outperformed the other.

In the SAW method, results showed that in 99.28 % of iterations, Tamanrasset had a higher score than Adrar. The histogram was strongly skewed positive, confirming that Tamanrasset consistently dominated across nearly all weight scenarios.

In TOPSIS, Tamanrasset's superiority was even stronger: in 99.84 % of iterations, its score exceeded Adrar's. The histogram displayed a clear positive divergence with negligible negative dispersion, highlighting TOPSIS's stronger decisiveness in recognizing Tamanrasset's structural advantage.

Overall, Monte Carlo analysis confirms that Tamanrasset remains the superior option under virtually all weight combinations, demonstrating high decision stability under uncertainty. This strengthens the credibility of the solution and adds a critical probabilistic dimension to MCDM, supporting risk- and uncertainty-aware decision frameworks.

7. Future works

In this section, it is proposed that future works incorporate other planning techniques that directly model system uncertainties. These techniques include the following:

- Two-stage Stochastic Programming for jointly modeling uncertainty in energy resources and demand [54];
- Scenario Aggregation Technique for the design of solar systems under uncertain climatic conditions [55];
- Modified BAVOA, combined with ARMA-based forecasting, for managing wind uncertainty and unit commitment planning [56].
- Future studies should examine the operational aspects of implementing the proposed systems. These include grid infrastructure needs, reliability under the harsh climate of southern Algeria, and the costs and challenges of maintenance in sparsely populated areas.
- To further develop the applied framework of this study, future research may consider employing metaheuristic optimization algorithms to determine the optimal configurations of hybrid energy

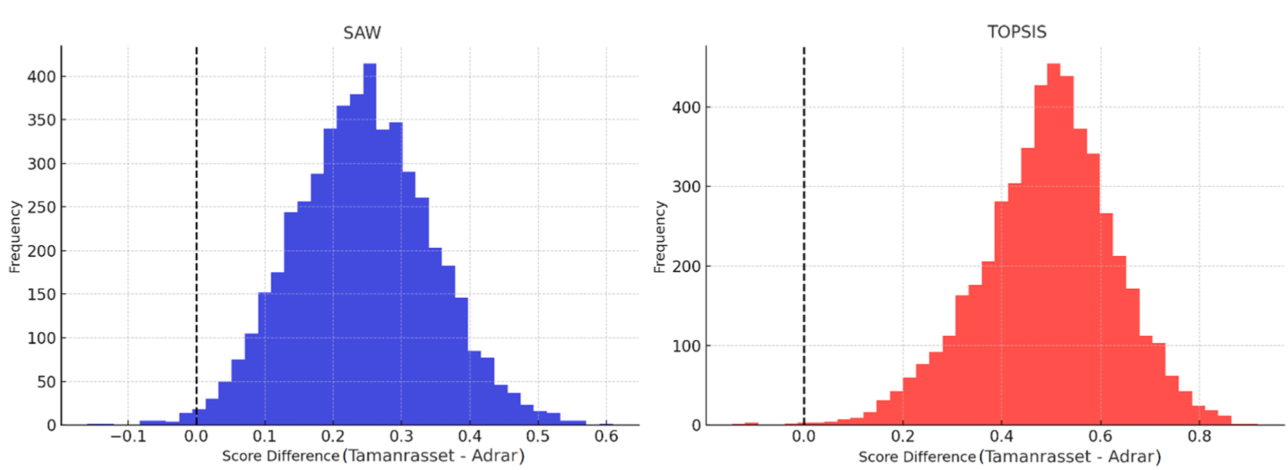


Fig. 11. Distribution of pairwise score differences (Tamanrasset – Adrar) based on 5000 Monte Carlo simulations for SAW (left) and TOPSIS (right). The x-axis shows the score difference per simulation, and the y-axis indicates the frequency of occurrences. A predominantly positive distribution confirms the robustness of Tamanrasset as the consistently superior option under random weighting scenarios.

systems and compare the outcomes with HOMER-based results. For instance, studies involving the Harmony Search and Firefly Algorithm for system design [57], or economic scenario analyses using various renewable resource combinations [58], may serve as appropriate references for integrating advanced optimization techniques into MCDM-based frameworks. These approaches can support more comprehensive exploration of the solution space and validation of system optimization under diverse conditions.
- Future research could complement HOMER-based modeling with metaheuristic optimization algorithms (e.g., Firefly, Harmony Search) to further expand the solution space and compare the robustness of outcomes across different methodological approaches [59,60].

It is also emphasized that implementing these techniques—especially in larger-scale systems or in countries with high variability in load, resources, or energy markets—can provide greater accuracy and flexibility for decision-making.

8. Discussion

The results of present work can be directly utilized to support Algeria's renewable energy vision for 2030 [61]. Given the regional nature of the proposed model and the designed multi-criteria scenarios, this framework can be applied to optimize energy resource selection at the local level and align implementation strategies with national strategic goals. Furthermore, the analysis of decision-making stability under varying scenarios provides a valuable tool for designing resilient policies in the face of economic, social, and environmental uncertainties. It should be emphasized that while only two locations were considered, the integration of six policy-oriented scenarios effectively increases the decision space and provides a justifiable set of alternatives, aligning the study's focus with scenario weighting rather than simple site selection.

The analytical framework proposed in this paper is transferable to other African countries with similar climates (such as Mali, Chad, Mauritania, and Niger) [62]. By adapting the input data and policy scenarios, this model can serve as a decision-support tool at the regional scale to enhance the deployment of renewable energy resources.

9. Conclusion

This study proposes a novel, replicable framework that combines scenario-based MCDM (SAW and TOPSIS) with both gradual and Monte Carlo sensitivity analyses. The approach provides a comprehensive way to assess and rank hybrid renewable energy systems under multiple policy perspectives. Unlike previous studies that focused mainly on technical or economic dimensions, this work incorporates social criteria such as population coverage and quantifies decision robustness under weight uncertainty, making it a valuable tool for resilient and socially-aligned energy planning in Algeria. This research is motivated by two key factors. The first is the urgent need to replace fossil fuels with clean energy in the arid and semi-arid climates of North Africa. The second is the lack of a comprehensive framework that integrates economic, technical, environmental, and social dimensions in evaluating energy systems. To operationalize this framework, we applied seven key criteria—including initial cost, LCOE, payback time, renewable energy production, energy losses, CO_2 emissions, and population coverage—to two distinct regions in Algeria: Tamanrasset and Adrar. The technical-economic simulation of energy systems was carried out using HOMER v2.81 software, followed by the multi-criteria ranking of options through SAW and TOPSIS methods under six different weighting scenarios. To assess the robustness of the results, a sensitivity analysis was performed both gradually (incrementally from 0 to 1) and randomly (Monte Carlo simulation with 5000 iterations). The key findings of this study are summarized as follows:

- The different MCDM weighting scenarios demonstrated that the selection of the optimal option is highly sensitive to the priorities of policymakers, environmental advocates, and engineers.
- In five out of six decision-making scenarios, Tamanrasset was decisively selected as the top-ranked option, particularly under cost-based, environmental, equity-oriented, and policy-driven perspectives.
- Tamanrasset's fully solar-based, zero-emission configuration—with lower initial investment and a rapid payback period (0.29 years)—makes it an attractive choice for low-cost, sustainable projects.
- Although the hybrid system in Adrar delivered higher energy output and showed better technical performance, it was only preferable in performance-based scenarios due to its higher emissions, costs, and lower population coverage.
- Sensitivity analysis revealed that criteria such as population, payback period, and CO_2 emissions had significant impacts on the final rankings—especially under the TOPSIS method, which exhibited more polarized and divergent behavior.
- In both SAW and TOPSIS methods, the population criterion had the greatest influence in boosting Tamanrasset's ranking, making it the ideal option in human-centered scenarios.
- Monte Carlo analysis with 5000 iterations confirmed that Tamanrasset remained the top-ranked alternative in >99 % of random weighting cases, highlighting the robustness and credibility of the final decision.
- A comparison of the two methods showed that SAW produced a more linear and gradual response, while TOPSIS exhibited sharper transitions and more decisive shifts under higher criterion weights.
- Criteria such as LCOE and initial cost had a stabilizing effect, with their influence becoming prominent only under extreme weight assignments.
- Overall, if the focus is on social equity, sustainability, and cost efficiency, Tamanrasset is the optimal choice; whereas, if technical performance and energy output are prioritized, Adrar may occupy a higher rank.

CRediT authorship contribution statement

Behrang Moradi: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Mehdi Jahangiri:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration. **Abderrahmane Khechekhouche:** Resources, Methodology, Investigation. **Maliheh Sadat Hashemi Mehr:** Validation, Software, Resources. **Mohammadhossein Nahavandi:** Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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